

EDUCATION

Arizona State University, Tempe, Arizona

Master of Science in Computer (Thesis)

Aug 2023 - Present

GPA: 4.00/4.00

Relevant courses: Machine Vision, Natural Language Processing, Statistical Machine Learning, Topics in Generative AI, Foundations of Algorithms, Data Mining, Multimedia and Web Databases, Perception in Robotics

SASTRA University, Thanjavur, India

Bachelor of Technology in Electronics and Communication Engineering

Aug 2016 - May 2020

GPA: 8.62/10.00

EXPERIENCE

Graduate Research Assistant at Cognition & Intelligence Lab, ASU | Advisor: Dr Chitta Baral

Jun 2024 - Present

- Developing a benchmark suite for fine-grained temporal reasoning in VideoLLMs (e.g., Gemini 2.0, Qwen2-VL, VideoLLaVa), focusing on event sequencing, causality, and temporal localization to advance model evaluation.
- Curating and annotating video QA datasets using Allen's interval algebra to assess implicit event inference and cross-temporal dependencies in VideoLLMs.

Software Development Engineer at Tata Consultancy Services, Chennai, India

Aug 2020 - Aug 2023

- Collaborated with NielsenIQ to architect and deploy a scalable data distribution pipeline to process large-scale market retail data using PySpark, Azure Databricks, and Hadoop, achieving seamless cloud integration and reducing data processing latency by 40%.
- Developed ML analytics models to evaluate pricing and promotions using NielsenIQ's consumer panel and retail scanner data. Implemented demand forecasting for FMCG sales, driving a 15% revenue increase and 10% market share growth for key brands.
- Automated data processing pipeline, increasing throughput by 3x through efficient large-scale Oracle SQL to Hive migration. Integrated Apache Kafka for real-time data ingestion and Apache Airflow for orchestration, enhancing query performance by 60%.

Research Intern at Tata Consultancy Services, Chennai, India

Jan 2020 - Mar 2020

- Implemented and compared two passage-based QA models: ALBERT and an optimized QANet replacing fixed sinusoidal with relative positional encoding for better capturing long-term dependencies and added a sliding window cache technique to reduce token length. Fine-tuned it on SQuAD, achieving an F1 score of 88.64 and EM score of 79.28.
- Deployed the best-performing model on cloud using Flask and built a web-based QA system. Fine-tuned the model on company-proprietary documents, improving FAQ-based user query response time by ~25%.

PUBLICATION

Evaluating Multimodal Large Language Models across Distribution Shifts and Augmentations

CVPR 2024

- Benchmarked Multimodal Large Language Models (InstructBLIP, LLaVA, LLaMA-AdapterV2) on real VQAv2 and synthetic CLEVR datasets, analyzing 75K curated QA pairs and identifying up to a 30% performance drop on synthetic images and 23% reduction under visual perturbations like Gaussian noise and zoom blur.
- Identified significant linguistic reasoning gaps, demonstrating a 20% F1-score disparity in yes/no answers under adversarial logical augmentations and a 72% bias towards affirmative responses in complex binary questions.
- Automated evaluation using Zephyr-7B as a judge model to assess MLLM vulnerabilities, quantifying a 15-25% performance degradation in color-based tasks and a 35% error rate in multi-object scenes, enhancing scalability of robustness assessments.

SOFTWARE SKILLS

Languages: Python, C++, C, PL/SQL, MySQL, Java, MATLAB

Libraries: PyTorch, TensorFlow, Keras, OpenCV, Langchain, CUDA, ONNX, FastAPI, Django, Flask, Hadoop, Spark

Tools: MongoDB, PostgreSQL, DynamoDB, Neo4j, Git, AWS, GCP, MLFlow, Airflow, Jenkins, ROS, Docker, Kubernetes, Linux

FEATURED PROJECTS

Anti-aliasing NeRF for novel view synthesis of 3D scenes

- Implemented vanilla [NeRF](#) from scratch in Pytorch on an A100 GPU, yielding a 22.28 PSNR and a 0.92 SSIM on Lego data.
- Enhanced memory efficiency via lazy data loading and optimized ray batch processing with parallelism.
- Added adaptive rendering module via selective sampling through image segmentation using [FastSAM](#) on [TanksandTemples](#) data, resulting in finer rendering in areas of interest with a 6% decrease in training time.
- Enabled scale-aware learning by incorporating cone tracing and integrated positional encoding from [MipNeRF](#). This reduced model parameters by 47% and improved SSIM by 15% on 1/8th resolution images while eliminating aliasing artifacts.

Self-Supervised Vision Transformer using DINO for Robust Representation Learning

- Implemented a knowledge distillation framework with no labels (DINO) on a Vision Transformer (ViT-Small) backbone featuring 8 attention heads. Benchmarked the representation learning model on an 80-category ImageNet classification task using PyTorch, achieving a test accuracy of 64.32%.
- Developed a novel patch-pruning method using attention map sparsity, accelerating inference by 1.4X with no accuracy loss by removing 15% of low-salience patches and maintaining 83% feature similarity in representations.
- Optimized multi-crop training with dynamic resolution scaling (2 global, 4 local), reducing GPU memory by 35% and accelerating convergence 1.8X for enhanced feature diversity.